

Impacts of the U.S. Safe Drinking Water Act Arsenic Rule on Arsenic Occurrence[†]

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Abstract

This study provides the first retrospective analysis of the causal impacts of the 2001 US Safe Drinking Water Act (SDWA) Arsenic Rule on arsenic (As) occurrence in public drinking water systems. Using publicly available SDWA compliance data, we employ

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a difference-in-differences strategy that compares As concentrations before and after the promulgation of the Rule for treatment relative to control water systems, defining treatment based on whether a water system had ever sampled As concentrations above the Rule’s 10 micrograms per liter ($\mu\text{g}/\text{l}$) threshold *before* the Rule was published. We find that annual mean As concentrations in treatment systems fell by almost 5 $\mu\text{g}/\text{l}$ ($p < 0.01$) after 2001 relative to control systems, representing a decrease of about 38% compared to pre-treatment levels. In light of the Rule’s projected net costs at the time it was promulgated, assessment of the Rule’s realized impacts using actual measured As concentrations from a nationwide sample is an important first step in retrospective benefit-cost analysis.

Synopsis: We provide the first retrospective evaluation of the 2001 Arsenic Rule’s impacts on arsenic occurrence, finding significantly lower concentrations among affected public water systems.

Keywords: Water pollution, drinking water, arsenic, retrospective analysis, Safe Drinking Water Act

Introduction

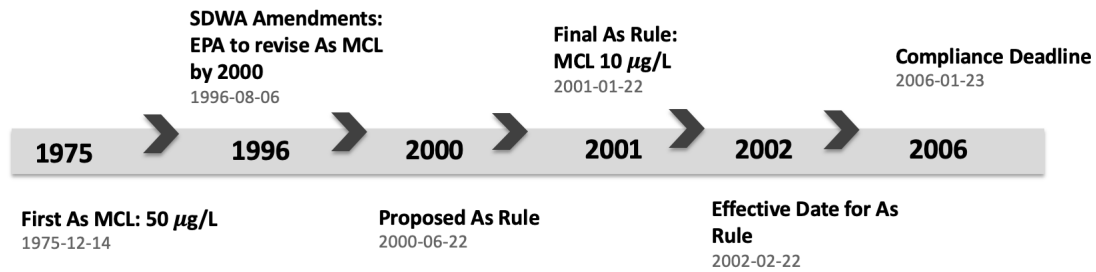
Arsenic (As) naturally occurs in the environment in soils, rocks, and volcanic materials. Some industrial and agricultural processes, for example those in glass and semiconductor manufacturing and the application of herbicides and pesticides¹, also contribute As to the environment. Inorganic As has implications for health: it is a known human carcinogen^{2–4}, and the literature documents associations between exposure to inorganic As and ischemic heart disease⁵, diabetes⁶, adverse pregnancy and birth outcomes⁷, and neurodevelopmental effects⁸. Due to these and other human health risks, the concentration of As in drinking water is regulated in many countries. The World Health Organization (WHO) established a guideline of 10 micrograms per liter ($\mu\text{g}/\text{l}$) for drinking water concentrations in 1993. As

of 2022, 136 countries had national drinking water regulations with a standard less than or equal to this WHO guideline, 40 had standards exceeding it, and 19 had no easily identifiable standard.⁹

In the United States, As in drinking water is regulated under the Safe Drinking Water Act (SDWA). On January 22, 2001, the U.S. Environmental Protection Agency (EPA) promulgated the Arsenic Rule, which lowered the maximum contaminant level (MCL) for As from 50 to 10 $\mu\text{g}/\text{l}$ (matching the WHO guideline), with compliance required by January 23, 2006.¹⁰ The Rule also implemented changes in compliance monitoring for As. Prior to the Rule, a system found in violation of the (weaker) 50 $\mu\text{g}/\text{l}$ standard was required to provide three additional samples within one month of the violation. After two consecutive samples below the 50 $\mu\text{g}/\text{l}$ MCL, the system returned to the default sampling frequency of at least one sample every three years for groundwater systems and annually for surface water systems. Under the Arsenic Rule, a violation of the 10 $\mu\text{g}/\text{l}$ MCL triggers quarterly monitoring. After a minimum number of samples, the state regulator can determine when the violating system returns to the default sampling frequency. The SDWA allows states to implement standards that are more stringent than federal MCLs, and two states, New Hampshire and New Jersey, have done so for As since 2001, an issue we return to later in the paper.

The Arsenic Rule has been controversial, in part because its projected economic benefits were less than projected costs at the time it was promulgated in 2001¹¹, and in part because of the potential for high compliance costs for systems serving small communities.^{12,13} Following the controversy, EPA postponed the effective date of the Rule by a year, to February 22, 2002, during which time EPA requested additional public comment and formed expert panels to reassess the new standard. Figure 1 presents a timeline of events associated with the Rule.

Figure 1: Arsenic (As) Rule Timeline



Notes: Figure 1 depicts a timeline of events relevant to the regulation of arsenic (As) in drinking water in the United States. In 1975, the National Interim Primary Drinking Water Regulations established the first enforceable As maximum contaminant level (MCL) of 50 µg/l. The Safe Drinking Water Act (SDWA) Amendments of 1996 directed EPA to propose a new As MCL by January 1, 2000. The initial proposed As Rule considered a standard of 5 µg/l but the final Rule established the new standard at 10 µg/l. Most water systems had to fully comply with the standard by January 23, 2006.

While the SDWA requires rigorous prospective benefit-cost analysis of new and revised MCLs, nothing in the statute requires retrospective review. Retrospective analysis is needed, however, to assess the causal impact of a policy on the response of regulated entities (in this case, public and private drinking water providers) once a Rule is passed, as well as the health impacts and other effects of that response.¹⁴⁻¹⁶ Since the Arsenic Rule's passage, research has explored some of its potential effects. For example, Foster et al.¹⁷ show that violations of the As standard initially climbed between 2006, the first year compliance with the new 10 µg/l standard was required, and 2008, and have declined steadily since then. They also show that most of the increased violations occurred in small systems and those using groundwater rather than surface water as a primary source. However, because the standard became more stringent, violations remain higher than they were prior to the Rule¹⁷, and it can be difficult to know how As occurrence and exposure have changed when considering changes in violations. Examining about 300 water systems in California, Grooms¹⁸ shows that As concentrations in drinking water may have declined as a result of the 2001 Rule, but that violations do not result in subsequent reductions in As concentrations among violating systems. The As Rule may also be associated with improved birth outcomes in California.¹⁹ In the epidemiological literature, Nigra et al.²⁰ analyze urinary As concentrations in the United States, comparing levels in 2003-2004 (after the Rule was promulgated, but before compli-

ance was required) with those in 2013-14. The authors show a decline in dimethylarsinate and total As concentrations between these two periods in nationally-representative samples (though of different individuals) among those who self-report access to public drinking water, but not among those using private water wells. While not a causal analysis, the estimated associations are consistent with the Arsenic Rule having lowered exposure to As in public drinking water in the United States.

Given controversy over the Arsenic Rule’s benefits and costs, uncertainty over the Rule’s impacts on intended outcomes, and the recent release of a final IRIS Toxicological Review of Inorganic Arsenic by the EPA¹, an important input to future rulemaking under the SDWA, additional work is needed to understand the causal impact of the Arsenic Rule on As occurrence and exposure. The literature evaluating the Arsenic Rule has been thin in part because historically the only available data for assessing regulatory impacts were violations data. The violations data identify when a water system exceeds an MCL, but not by how much. It is also impossible to observe any changes in As concentrations below the MCL in the violations data. Thus, violations are a noisy and incomplete measure of drinking water quality. Prior work assesses the impact on As occurrence of a more stringent drinking water standard in Denmark²¹, but there has been no such assessment of the 2001 Arsenic Rule in the United States.

As an essential first step toward a retrospective quantification of the Arsenic Rule’s benefits, we use publicly-available data capturing spatial and intertemporal variation in As concentrations in drinking water to estimate the Rule’s impact on arsenic occurrence in community water systems. We assemble SDWA compliance samples from 1995 to 2019 using occurrence information collected for the Rule itself as well as three Six-Year Reviews of Drinking Water Standards by the EPA.^{22–25} An advantage of using compliance samples is that one can create a panel of As sampling records for systems before and after the Rule using *actual* As concentrations for regulated water systems as opposed to numbers of violations, which may mask important variation in As occurrence. The SDWA compliance samples are

routinely collected by states and used for monitoring and enforcement, but they have only recently been employed by researchers.²⁶⁻²⁸

Materials & Methods

Arsenic Occurrence Data

Public drinking water systems in the United States serve about 90% of the population, with a majority of people receiving drinking water from community water systems (CWS).^{29,30} CWS are public water systems that serve at least 25 people or at least 15 connections year-round. To estimate the impacts of the Arsenic Rule, we use SDWA compliance samples for CWSs, assembled from four sources: the occurrence analysis for the Rule itself, the Six Year Review 2, the Six Year Review 3, and the Six Year Review 4.²²⁻²⁵ These data include sample information from 24, 45, 51 (including Washington, DC) and 48 states over the periods 1980-1997, 1998-2005, 2006-2011, and 2012-2019, respectively. The Six Year Review (SYR) datasets request all compliance monitoring information from state drinking water regulatory authorities, including samples from community water systems, non-community water systems, and across all system size categories. These records are then used to determine whether any standards merit revision. Because the SYR information collection request asks for existing data, there is no additional burden to producing the records for smaller systems and thus we do not expect selection on system characteristics in SYR records. That said, some states have not reported any information to the SYR, and hence those states are absent from this analysis.^{23,24,31}

We focus on water systems consistently classified as CWS during this period because CWS were required to meet the new, more stringent standard, and because we observe As samples for many CWS both before and after the Rule, making it possible to estimate the Rule's impact from comparisons of As concentrations over time. Reductions in As occurrence likely also occurred among non-transient non-community (NTNC) water systems, a smaller

class of public water system. These systems had been excluded from the old 50 $\mu\text{g}/\text{l}$ standard but were required to meet the more stringent 10 $\mu\text{g}/\text{l}$ standard under the Rule. However, NTNC water systems were not required to monitor As prior to the Rule, leaving us little data to identify the Rule’s impact on As occurrence for these systems. This means that our estimates of the causal effects of the Rule on CWS will be partial estimates of the Rule’s total effects on As in drinking water.

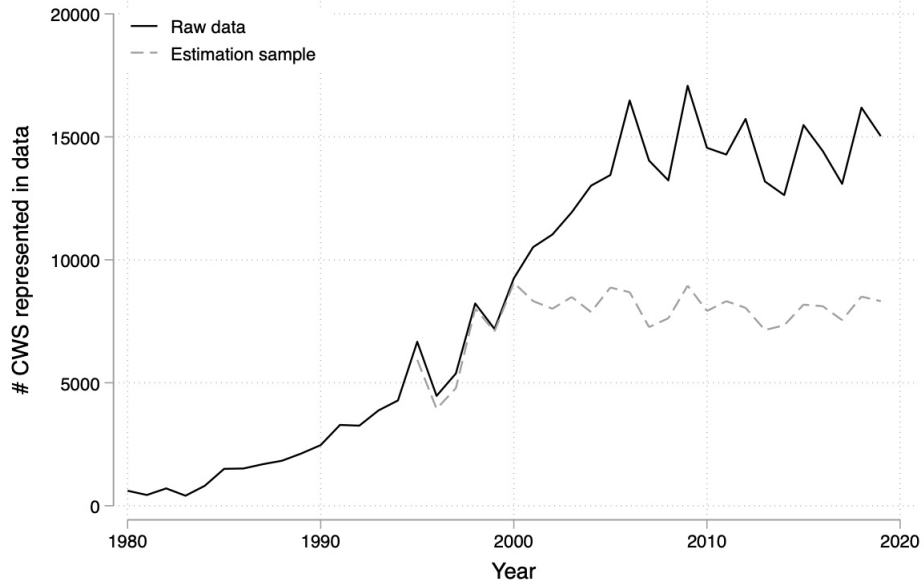
Table S1 in the Supporting Information (SI) compares mean characteristics of systems that are included in the study sample with those of all active CWS that are not in our study from the Safe Drinking Water Information System (SDWIS) as of 2011. System characteristics for comparison include system size, water source, and geographic region defined according to the US Census. Among CWS in the study sample, a greater share are higher capacity systems and a smaller share are very small systems. Sample CWS are also more likely than the full CWS population to source from groundwater. In terms of geographic representativeness, compared to non-study CWS, systems included in our study are more likely to be located in the West and Midwest and less likely to be in the Northeast and South. The differences in water source and region are similar when focusing on the sample of small or very small CWS only. While higher capacity systems are more likely to participate in the Six Year Reviews, it is unclear how the difference in composition would affect estimated impacts of the As Rule. If the smaller systems that participated are disproportionately those with more resources and better treatment systems, then the impacts we estimate are likely to be a lower bound on the true effect. On the other hand, if those that participated are the most troubled systems, then we might over-state the policy impact.

The compliance samples in our data provide actual As concentrations for regulated water systems collected at each entry point to the distribution system, allowing us to create a panel of As sampling records for CWS before and after the Rule. If As is not detected, then we assume a value of zero in these cases and assess sensitivity to the possibility that As was present below the detection limit in the Results & discussion section. Because the majority

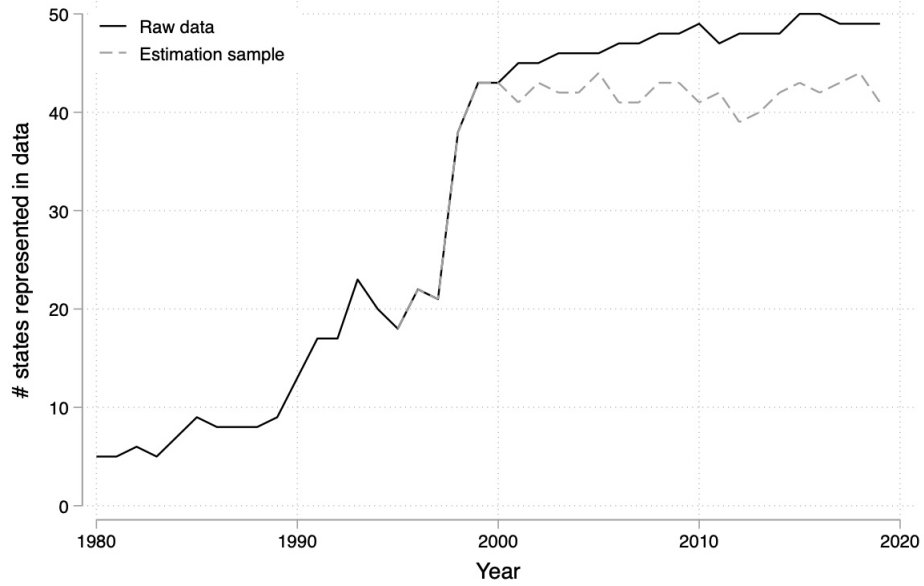
of the samples in the study use an As detection method called Method 200.8, which has a minimum detection level of $1.4 \mu\text{g}/\text{l}$, we format all output to include one decimal point. The sampling records also include information on the time and location of the As sample, the sample protocol such as the method detection limit, and details regarding the water system such as county location and whether the CWS uses groundwater or surface water as a source.

For most of our records, we also observe whether a sample was collected from an untreated or treated sampling point (i.e., “raw” vs. “finished” water), but we lack this information from 1998-2005. We do not exclude raw water samples to avoid introducing inconsistencies in the data over time that could confound effects of the change in the MCL. We select 1995 as the start of our sample period given that fewer systems and states are represented in the data in the 1980s and early 1990s (Figure 2).

Figure 2: Data Coverage of Systems and States Over Time



(a) Number of unique community water systems



(b) Number of unique states

Notes: Panels (a) and (b) respectively show the number of unique community water systems (CWS) and unique states represented in the raw data each year from 1980 to 2019, using a solid line. In both panels, the dotted line captures the same information, but for the sample we construct from the raw data (which spans 1995-2019) to estimate the impacts of the Arsenic Rule on arsenic occurrence.

From these raw data, we construct a sample that reflects CWS with at least one As sample during the pre-Rule period (1995-2000) and at least one sample during the post-Rule period (2001-2019). We then aggregate the data to the system-by-year level. Aggregating the data to the system-year level yields the mean As concentration for each system i and year t provided the data contain at least one As concentration for system i in year t . Our final sample is an unbalanced system-by-year panel reflecting 22,127 community water systems over 25 years.

Empirical Methods

Methods: differences in means

To assess the impact of the As Rule on As occurrence among sample CWS, we first simply compare differences in raw mean concentrations among systems before vs. after the Rule is implemented. However, this approach does not provide an estimate of the Rule’s causal effect on concentrations, because there are many potential confounders (e.g., changes in the natural environment, changes in anthropogenic As releases, changes in economic activity) that might explain any observed differences. Obtaining causal effects of the Rule on As concentrations requires a research design that removes these kinds of concurrent trends in As concentrations from the estimated impact of the policy.

Methods: differences-in-differences

To this end, we employ a difference-in-differences (DD) strategy that defines treatment based on the 2001 Rule’s regulatory threshold of 10 $\mu\text{g/l}$. Specifically, we consider treated water systems as those where measured As levels were ever above the proposed regulatory threshold *before* the Rule was published in 2001. Control systems are those for which we never observe an As concentration above 10 $\mu\text{g/l}$ prior to 2001. We compare how As concentrations changed for the treated group relative to the control group before and after the Rule. Because the Rule is not binding for water systems with baseline As levels below the regulatory threshold

(i.e., control systems), changes in As in these systems likely reflect correlated changes in As from the environment or economic activity rather than changes due to any mitigation measures.

We estimate the baseline DD regression specification in equation (1).

$$Y_{it} = \beta_0 + \beta_1 \text{aboveMCL}_{it}^{2001} + \gamma_i + \delta_t + u_{it} \quad (1)$$

Y_{it} represents the mean As concentration for system i in year t . The variable $\text{aboveMCL}_{it}^{2001}$ is equal to 1 for treatment systems in post-treatment years (e.g., after 2000) and 0 otherwise. γ_i and δ_t respectively represent water system and year fixed effects. γ_i controls for time-invariant water system characteristics (e.g., geographic location), and δ_t controls for time-varying factors common to all water systems (e.g., national macroeconomic trends). β_1 represents the DD estimate, the coefficient of interest, which measures how As concentrations for treated systems changed on average relative to control systems following treatment. Under the assumption that the set of control systems reflect how As concentrations would have changed for the treated systems in absence of the Rule (i.e., parallel trends), β_1 is an estimate of the causal impact of the Rule on As concentrations in CWS.

Using the DD approach, we also explore three different dimensions of heterogeneity based on observable system characteristics: system size (service population), region, and water source (groundwater vs. surface water). We re-estimate equation 1 on different subsamples defined by these characteristics, to obtain different estimates of β_1 . Note, however, that these system characteristics are not independent. For example, very small (serving fewer than 500 people), small (serving between 501 and 3,300 people), and medium (serving 3,301 and 10,000 people) systems are more likely to use groundwater sources exclusively compared to large (serving 10,001 and 100,000 people) and very large systems (serving more than 100,000 people), which are more likely to have some surface water sources.

Methods: event study

Using the same basic intuition (and the same definition of treatment and control CWS) as our DD approach, we also estimate an event-study specification, as in equation (2), to illustrate the dynamic effects of the Rule on As occurrence:

$$Y_{it} = \beta_0 + \sum_{t \in (1995, 2019), t \neq 2000} \alpha_t \cdot treated_i \cdot D_t + \gamma_i + \delta_t + u_{it} \quad (2)$$

In equation (2), $treated_i$, which takes the value of one if system i is a treatment system and zero otherwise, is interacted with a set of indicator variables denoting event time. We express event time in calendar years with 2000, the year prior to the final As Rule’s promulgation, as the reference year. The coefficients of interest are the estimates of α_t , which reflects the average difference in mean annual As occurrence between treatment and control systems in year t compared to 2000. Under the assumption of parallel trends, the estimated α_t coefficients in the post-treatment period capture the causal effects of the Rule on As occurrence as they unfold over time. In the pre-treatment period (i.e., before 2001), the α_t are essentially placebo tests, such that statistically non-zero estimates of α_t prior to the Rule would threaten the assumption of parallel trends in As occurrence between control and treatment CWS absent the Rule. The remaining parameters in equation (2) are identical to those in equation (1).

In our event study analysis, a potential threat to identification arises from increased testing requirements under the Arsenic Rule, which mandate that CWS exceeding the MCL must test more frequently and publicly report results. This increased scrutiny and testing frequency could lead to the detection of higher arsenic levels in systems we initially classified as controls. Such factors could cause violations of the parallel trends assumption required for a causal interpretation of our DD estimates. To assess the implications of potential violations of parallel trends, we apply the “honest” difference-in-differences (HonestDiD) approach proposed by Rambachan and Roth³² to our event study estimates. HonestDiD is

designed to statistically extrapolate any pre-treatment trends to the post-treatment period under controlled deviations, bounded by how much treated and control group trends differ before treatment. By doing this, it examines the sensitivity of our estimated treatment effects to any such differences. That is, the approach can reveal how strongly the parallel trends assumption must hold in order to interpret our event-study estimates as causal.

Coefficients and standard errors for all DD and event study models reported in the tables and figures are estimated in Stata 17, using `reghdfe`. Results for the HonestDiD approach are estimated in Stata 18, using `honestdid`.

Results and discussion

Creation of treatment and control groups

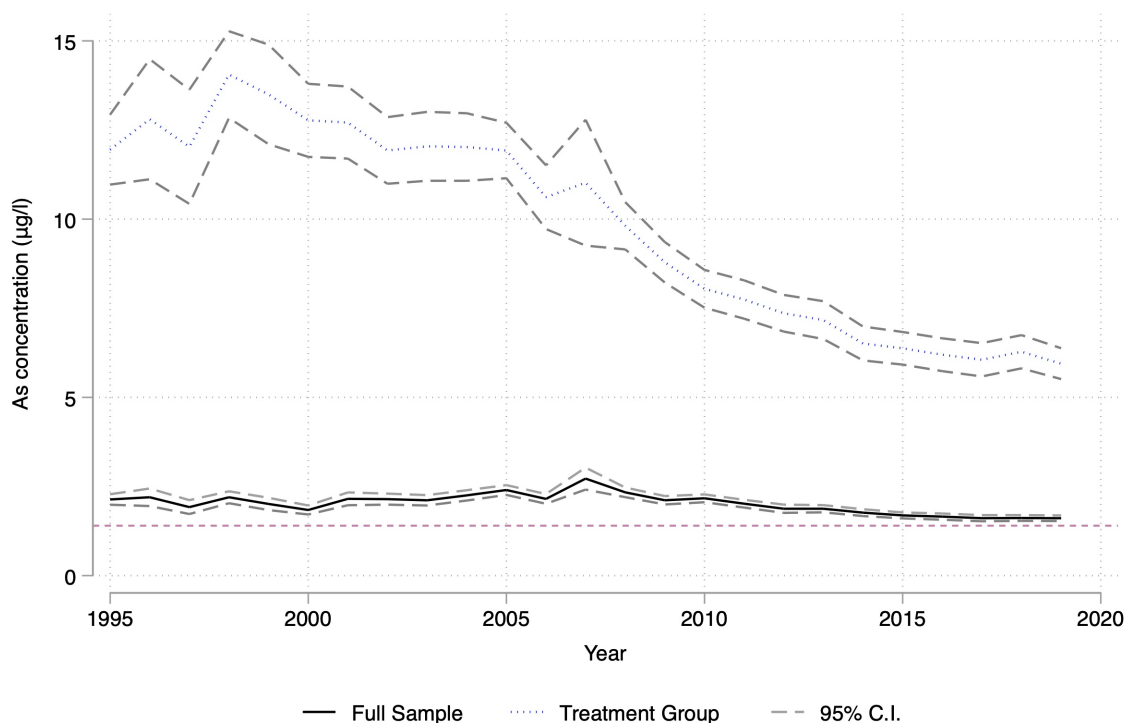
The solid line in Figure 3 depicts annual averages of mean As concentrations for sample CWS over the sample period from 1995-2019 as well as 95% confidence intervals. From 1995 to the early 2000s, annual averages of mean As concentrations for sample CWS fluctuated around 2 $\mu\text{g/l}$, but then there is a notable spike around 2007. After the spike, there is a downward trend in annual averages of mean As concentrations for sample systems through the end of the sample period.

The increase observed around the 2006 compliance date is likely due to compositional changes. The initial monitoring period under the stricter standard was 2005-2007. Groundwater systems, which must sample every three years, had to complete sampling by December 31, 2007; the sampling deadline for surface water systems, which are required to sample annually, was December 31, 2006.³³ Thus, the spike may have resulted from systems with higher average As concentrations delaying sampling until just prior to the initial sampling deadline. Additionally, systems with higher As concentrations were subject to more frequent testing under the new rule, which may have further increased the reported average As level during this period. Finally, the new rule required more water systems to report As levels,

including those that previously may not have been regularly testing or publicly reporting their results. (This change in the number of systems reporting over time can be seen clearly in the raw data in Figure 2).

In general, the annual averages of mean As concentrations mask considerable heterogeneity in sampling values within and across sample water systems. The overall trends shown by the solid line in Figure 3 are driven by As concentrations in ground water-based systems, which constitute 87% of the sample (see Figure S1). In addition, while the annual mean As concentrations for the *average* system in our sample over the period from 1995 to early 2000s were well below 10 $\mu\text{g}/\text{l}$, there were certainly exceedances of this threshold during the period before the compliance deadline in 2006. Perhaps unsurprisingly, the annual averages of mean As for CWS that ever exceeded 10 $\mu\text{g}/\text{l}$ before 2001 (the treatment group), shown by the dotted line in Figure 3, consistently exceeded the 10 $\mu\text{g}/\text{l}$ during this period.

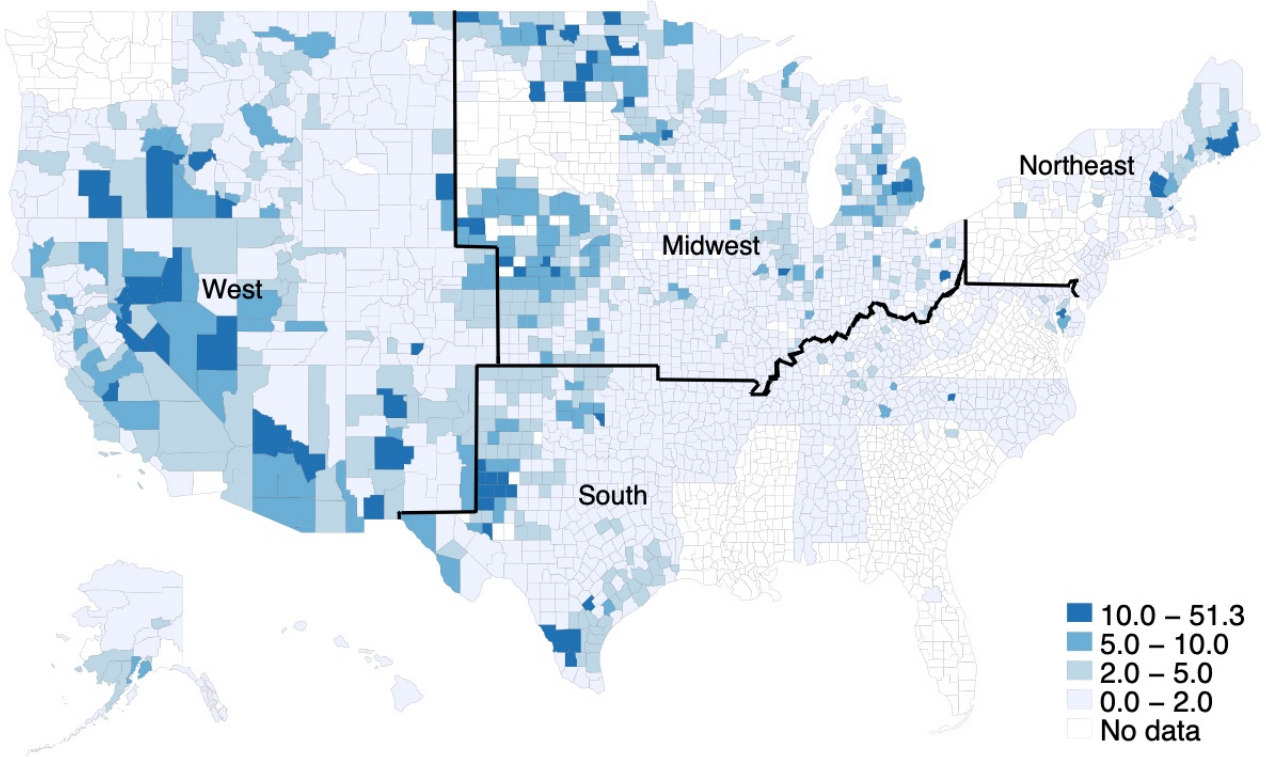
Figure 3: Trends in Average Annual Mean As Concentrations among Sample and Treatment Community Water Systems, 1995 to 2019



Notes: The figure graphs the annual average mean arsenic (As) concentration (and 95% confidence intervals) in $\mu\text{g/l}$ for the 22,127 community water systems (CWS) in our sample from 1995-2019 (solid line) as well as for treatment systems (dotted line), CWS that ever sampled As greater than $10\mu\text{g/l}$ before 2001. The most common detection limit of $1.4\mu\text{g/l}$ is denoted as the horizontal line. The panel is unbalanced, because we do not observe As concentrations for each system in every year.

To depict spatial variation in baseline As concentrations among sample CWS, we aggregate the data at the county level. Figure 4 shows the average annual mean As concentration by county over the period 1995-2000, prior to the Rule. Systems for which we do not observe a geographic location, about 13.5% of the sample, are not reflected in the figure, and counties shaded white have no data. Counties in the southeastern US that are represented in our sample generally had lower As concentrations compared to other areas prior to the Arsenic Rule. As concentrations were generally larger in the West in the pre-Rule period.

Figure 4: Average Annual Mean As Concentration by County, 1995-2000



Notes: The figure shows the average annual mean As concentration among community water systems (CWS) by county in the pre-Rule period, 1995-2000. Counties are grouped into 5 categories by average concentration in $\mu\text{g/l}$: 0.0-2.0, 2.0-5.0, 5.0-10.0, and 10.0-51.3 (those in white have no As data). The first group includes counties in which the average annual mean As concentration among CWS in the pre-Rule period are far below any US maximum contaminant level (MCL) for As. The second group shows counties with CWS that are close to the more stringent standard of 5 $\mu\text{g/l}$ adopted by New Jersey and New Hampshire. The third group reflects counties with CWS just below the 10 $\mu\text{g/l}$ standard adopted by the Rule, and the fourth group includes counties in which the average annual mean As concentration among CWS in the pre-Rule period exceeds the 10 $\mu\text{g/l}$ MCL. The figure was produced in Stata 18 using `maptile`.

Almost 9% of systems in our sample (1952 systems) are assigned to treatment with the remaining systems serving as controls. Table S2 compares average treatment and control systems based on observable system characteristics in the pretreatment period (i.e., prior to 2001). Compared to control systems, treatment systems are on average more likely to be very large (serving more than 100,000 people) or large (serving between 10,001 and 100,000 people) but less likely to be very small (serving fewer than 500 people). In addition, the average treatment system is more likely than the average control system to exclusively source groundwater. Finally, relative to control systems treatment systems are less likely to be located in the South and Midwest and more likely to be located in the West. These level mean differences in observable system characteristics between treatment and control systems

do not threaten a causal interpretation of our DD and event study results.

Differences in means

Table 1 provides averages of annual mean As concentrations for treatment and control systems before and after the Rule’s promulgation in 2001. The averages of annual mean As concentrations for control systems before and after 2001 were respectively 0.7 and 1.0 $\mu\text{g/l}$, well below the 10 $\mu\text{g/l}$ standard. In contrast, the average of annual mean As concentrations for treatment systems fell from 13.0 $\mu\text{g/l}$ before 2001 to 8.8 $\mu\text{g/l}$ after the Rule was passed. Viewed differently, annual mean As concentrations for treatment systems were, on average, 12.3 $\mu\text{g/l}$ higher than control systems before 2001, but only 7.8 $\mu\text{g/l}$ higher, on average, after the Rule. These preliminary differences in raw means suggest that the Rule could have decreased annual mean As concentrations for treatment systems by 4.5 $\mu\text{g/l}$ or about 35% relative to their baseline, pre-2001 As levels. As noted earlier, however, this difference in means cannot be interpreted as the causal effect of the Rule.

Table 1: Averages of Annual Mean As Concentrations

	Treatment	Control	Difference
	(1)	(2)	(1)-(2)
Before 2001	13.0	0.7	12.3
	(17.2)	(1.7)	[0.0]
After 2001	8.8	1.0	7.8
	(13.2)	(3.01)	[0.0]

Note: Table 1 reports means of annual average arsenic (As) concentrations in sample community water systems (CWS) before and after 2001 (inclusive of 2001) by treatment status. Standard deviations are in parentheses. The final column reports the difference in means by treatment status and, in brackets, the p-value associated with a t-test for equality of means. 1,952 unique CWS are assigned to the treatment group. 20,175 unique CWS are assigned to the control group.

Difference-in-difference estimates

The first column of Table 2 presents the DD estimate of the Rule’s impact (β_1 from equation 1). Standard errors are clustered at the water system level and are provided in parentheses. After 2000, annual mean As concentrations in treatment systems fell by 5.0 $\mu\text{g/l}$ ($p < 0.01$)

relative to control systems, representing a decrease of about 38% compared to pre-2001 levels, somewhat more than the raw mean difference from Table 1 would suggest. In the second column of Table 2, we show robustness to an alternative definition of the post-treatment period: 1997 and later, after the impending rulemaking had been announced but prior to resolution of uncertainty regarding specifics of the Rule. Between 1997 and 2001, some systems may have invested in control technologies in anticipation of the impending Rule. The results reported in column (2) explore this. Using the alternative treatment definition (i.e., 1997+), the DD estimate suggests an average reduction of 3.6 $\mu\text{g}/\text{l}$ in annual mean As concentrations among treatment systems compared to control systems, slightly lower than our main DD estimate. Results are qualitatively similar when we estimate a log-linear version of equation 1 (Table S3).

Table 2: Estimated Impact of the Rule on Annual Mean As Concentrations

	Post-treatment period	
	2001+	1997+
	(1)	(2)
β_1	-5.0*** (0.3)	-3.6*** (0.3)
Pre-treatment average	13.0	12.3
Estimated effect size	-38.5%	-29.3%

Notes: The table presents the estimated impact of the Rule on annual mean arsenic (As) concentrations during two post-treatment periods: 2001+ and 1997+. Standard errors are clustered at the water system level and are provided in parentheses. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sample size is 192,347 observations, representing 22,127 unique community water systems. Pre-treatment average is the average of annual mean As concentration for treatment systems in the pre-treatment period, as defined in the respective column.

How does the size of the estimated reduction in As based on our retrospective analysis compare to the results of the prospective Regulatory Impact Analysis (RIA) conducted by EPA? When projecting the likely benefits of the Rule in its *ex ante* RIA, EPA assumed that affected systems would reduce As concentrations to 0.8 times the new 10 $\mu\text{g}/\text{l}$ standard (or to 8 $\mu\text{g}/\text{l}$) on average.³⁴ To produce a post-Rule As concentration based on our retrospective analysis that is appropriate for comparison, we subtract an estimated value of β_1 in Table 2 from the relevant pre-treatment average. Using the 2001 post-treatment period definition,

the estimated post-Rule average annual mean As concentration for treated CWS based on our DD estimate is 8.0 $\mu\text{g/l}$, remarkably similar to the prospective RIA’s estimate. We obtain a slightly higher estimate of 8.7 $\mu\text{g/l}$ when we use the alternative post-treatment period definition (i.e., 1997+).

Robustness of baseline DD estimates

We test the robustness of our main DD results in several ways. First, we examine the possibility that changes in state As standards for drinking water may be responsible for the effects we estimate. During our sample period, two states implemented a more stringent 5 $\mu\text{g/l}$ standard: New Hampshire (July 12, 2019, with compliance required by July 1, 2021) and New Jersey (Nov. 4, 2004, with compliance required by Jan. 23, 2006). The 10 $\mu\text{g/l}$ federal standard was binding only in the remaining 48 states after these standards took effect. In Table S4, we re-estimate our main specification excluding all CWS in New Hampshire and New Jersey. The estimates are very similar to our baseline estimate.

Second, we test for potential bias that could result from the fact that the compliance samples in our data are self-reported by CWS, and the literature identifies strategic sampling as a challenge for monitoring and enforcement under some SDWA rules, particularly those in which violations occur once the number of problematic samples exceeds a percentage threshold.^{35,36} Some aspects of the As Rule use percentage thresholds to determine compliance, and if CWS are sampling strategically under the Rule, such results could be included in our data. If this behavior increased for treated systems starting in 2001 and it resulted in additional lower-than-MCL samples for treated systems, our approach could over-estimate the Rule’s impact on As occurrence. We investigate the sensitivity of our estimates to this issue by restricting the composition of our systems to those with “regular sampling,” assuming a three-year compliance cycle, and systems that remain active long after the Rule is first implemented. Specifically, we re-estimate our main DD model on a sample of water systems that have a minimum of one sample every three years for the post-treatment period

(2001-2019) as well as for the entire study period (1995-2019), and we limit the sample to systems that were still active in 2015. Results are presented in Table S5. These restrictions on sampling frequency do not meaningfully change our estimates, which are all slightly larger than our baseline estimates.

Third, we test for sensitivity to different assumptions regarding non-detects. Our main results assume zeros for non-detects. Table S6 reports estimates assuming, instead, that non-detect values are (i) 50% of the detection limit, (ii) at the detection limit, or (iii) $1.4 \mu\text{g/l}$ (the method detection limit for Method 200.8, which was used as the basis for setting the technically feasible threshold of the MCL^{37,38}). The samples for specifications (i) and (ii) are limited to systems that record detection limits. To assess whether changes in estimates are due to the assumption about non-detects or sample differences from missing detection limit information, we also present results using our baseline non-detect value of zero but with the sample that assumes non-detects are at the detection limit. Compared to the baseline estimate (column 1) imputing non-detects based on either half of the detection limit (column 2) or at the detection limit (column 3) increases the treatment effect to -6.9 and -8.9, respectively. The differences in estimated impacts are not driven by changes in the sample: the treatment effect using a value of zero based on a sample with non-missing detection limit information is -4.6 (column 4). The estimated impact when using a $1.4 \mu\text{g/l}$ detection limit (column 5) is similar to the baseline estimate. We thus conclude that the choice of zero for non-detects may tend to understate the impact of the policy.

Fourth, we consider the implications of our inability to distinguish between “raw” and “finished” water samples for part of the sample period (1998-2005), leading us to retain raw water samples for all years. Raw water tends to have higher contaminant levels than finished water, so this decision could result in higher As levels than were actually present in tap water. To test the sensitivity of the main result to this concern, we re-estimate the main DD specification based on a sample of finished water only. We create this sample by either dropping identifiable raw water samples or keeping identifiable finished water samples. We

also turn to the occurrence data to capture sampling between 1998-2000 instead of relying on SYR2 data for 1998-2000, which does not have sampling type information. As reported in Table S7, the estimated impacts on finished water range from -5.5 to -6.2 $\mu\text{g}/\text{l}$, similar to our main estimate of -5.0 $\mu\text{g}/\text{l}$ under the 2001+ post-treatment period. If anything, our choice appears to have been conservative. It is also noteworthy that the quality of raw water improves (-1.8 $\mu\text{g}/\text{l}$, $p < 0.01$), suggesting that some water systems may have switched water sources as a means of complying with the Rule.

Variation in DD effects related to system characteristics

The DD results provide estimates of the average effect of the Rule on annual mean As concentrations for community water systems. While an estimate of the average effect is useful, if different types of CWS face different costs of compliance with the stricter MCL, then we might expect to see variation in the Rule's effects across systems. Indeed, EPA's prospective RIA of the Rule estimated variation in compliance costs across systems based on treatment technology, system size, and source water, among other factors.^{34,39}

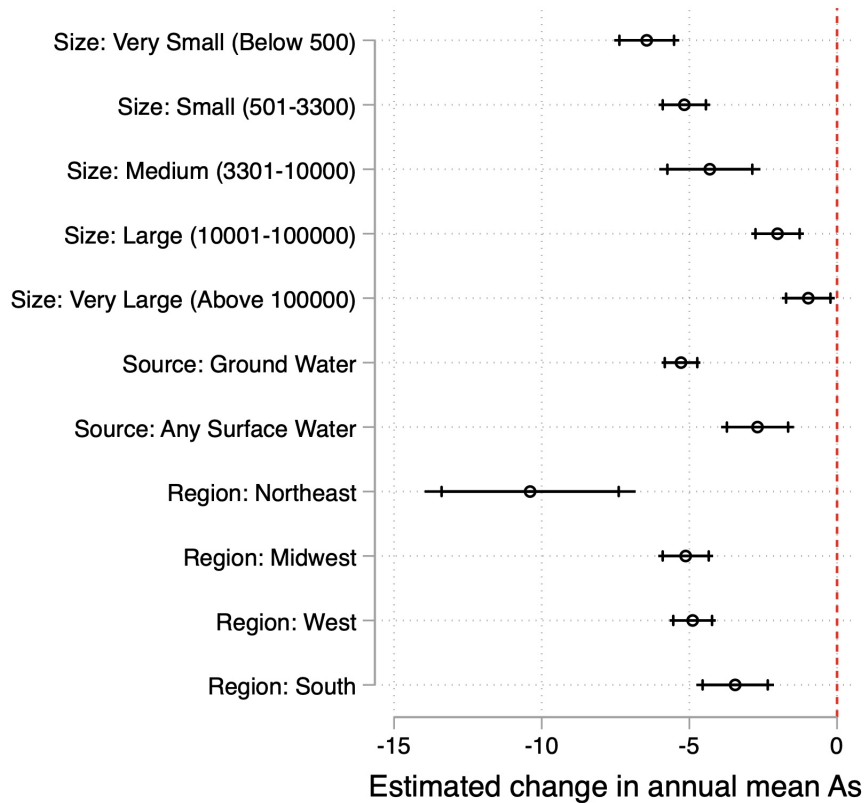
Figure 5 graphs coefficient estimates and 95% confidence intervals from the tests for heterogeneous effects. Each estimate in the figure is from a different regression in which we estimate β_1 from equation (1) for different sub-samples defined based on observed system characteristics: system size, water source (groundwater vs. surface water), and geographic region.

Consider, first, the results with respect to system size. Consistent with the *ex ante* cost estimates in EPA's prospective RIA, the estimated reductions in As concentrations from the Rule occur primarily in smaller CWS, especially those serving fewer than 10,000 people. The estimated β_1 for very small systems is -6.4 $\mu\text{g}/\text{l}$, corresponding to about a 38% reduction when evaluated at the pre-treatment subsample mean for very small treatment systems of 16.8 $\mu\text{g}/\text{l}$. In contrast, the estimated β_1 for large systems is -2.0 $\mu\text{g}/\text{l}$, about a 29% reduction when evaluated at the pre-treatment subsample mean for large treatment systems of 6.8 $\mu\text{g}/\text{l}$.

That smaller water systems saw the largest impacts of the Rule is consistent with concerns over high compliance costs for small systems. Note, however, that the larger estimated reduction for small systems is also suggestive of larger benefits for communities served by smaller systems.

We also estimate different effects of the Rule based on source water. The estimated reduction for groundwater-sourced systems is almost twice that of systems that source some surface water. Lastly, we explore regional variation in the Rule’s estimated impact. Figure 5 shows similar estimated effects for systems located in the Midwest and West. Compared to these regions, estimated reductions are slightly smaller for systems in the South and largest for systems in the Northeast.

Figure 5: Heterogeneous Effects by System Size, Water Source and Region

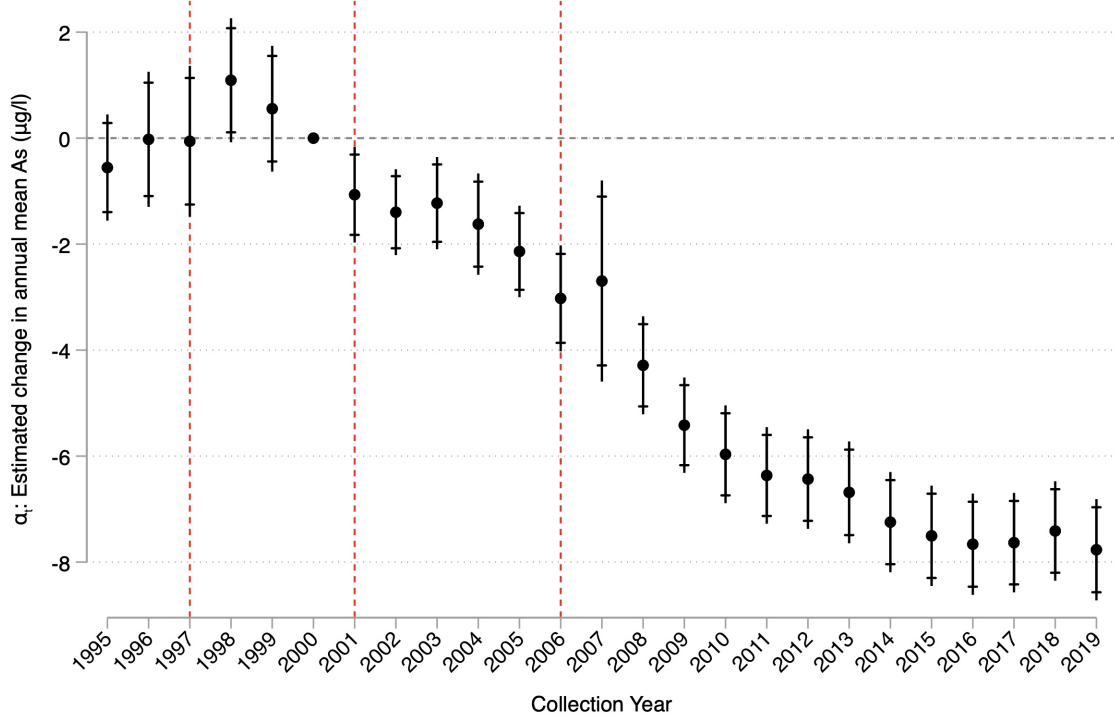


Notes: Figure 5 reports estimates of β_1 and associated 95% confidence intervals from equation (1) using subsamples based on observed water system characteristics. Each estimate in the figure comes from a different subsample regression reflecting the restriction described on the y-axis. Subsamples by geography correspond to the four U.S. Census regions: Northeast, Midwest, West, and South. See Figure 4 for region boundaries.

Event study estimates

Figure 6 reports results from estimating the event study model in equation (2), where we recover an estimated impact of the Rule for each post-treatment year relative to 2000. The estimated coefficient for year t reflects the average difference in mean As occurrence between treatment and control systems in year t compared to 2000. Beginning in 2001, mean As concentrations among treatment systems show a statistically significant decrease compared to control systems, relative to the base year. These declines continue through the early 2010s when they start to level off. Figure 6 also shows limited evidence of differential trends in mean As concentrations between treatment and control systems before the Rule was passed, which strengthens the interpretation of our DD and event study results as reflecting the estimated *causal* change in As from the Rule. As demonstrated in the SI, results are similar with a log-linear specification (Figure S2), when we restrict our analysis to the states included in the original occurrence data used to support the Rule (Figure S3), and when we use 1997 instead of 2001 as the base year for comparison (Figure S4).

Figure 6: Event Study Estimates



Notes: The figure shows estimated coefficients (α_t from Eq. 2, or the coefficients on the variable $treated_i \cdot D_{it}$) and 95% confidence intervals from an event study specification with annual mean arsenic (As) concentrations as the dependent variable. Estimates of α_t represent the average difference in mean annual As occurrence between treatment and control systems in year t compared to 2000. The estimated α_t coefficients in the post-treatment period (2001-2019) capture the causal effects of the Rule on As occurrence as they unfold over time. In the pre-treatment period (pre-2001), the α_t are placebo tests; statistically non-zero estimates of α_t prior to the Rule would threaten the assumption of parallel trends in As occurrence between control and treatment water systems absent the Rule. Vertical lines in the figure represent the year before rule proposal (1996), year before rule promulgation (2000), and year of compliance (2006) with the rule.

Honest difference-in-difference estimates

Recall that a potential threat to identification arises from increased testing requirements under the Arsenic Rule, which mandate that CWS exceeding the MCL must test more frequently. This increased testing frequency could lead to the detection of higher As levels in systems we initially classified as controls and cause violations of the parallel trends assumption. In fact, we can see in Table 1 that control systems experienced a small increase in average annual mean As concentrations subsequent to the Rule, suggesting that control and treatment groups may have followed different trends even in the absence of the Rule.

Figures S5 and S6 in the SI present the results from the HonestDiD approach we im-

plement to test for sensitivity of our event-study results to violations of parallel trends. In these figures, $\bar{M}$ represents the magnitude of deviations from the parallel trends assumption permitted post-treatment. We apply the HonestDiD test separately to each post-treatment event study estimate from 2001-2019. Estimated treatment effects for 2001–2005, during the early Rule implementation period, are sensitive to potential violations of the parallel trends assumption. However, from about 2006 onward, the estimates become increasingly robust to such violations. Between 2009 and 2019, the estimated treatment effects remain statistically significant and of similar magnitude even when allowing for large post-treatment deviations up to $\bar{M} = 2$, meaning treatment and control group trends could diverge by up to 200% of the maximum deviation observed in the pre-treatment period. These findings suggest that our results for the later period, after the Rule’s full implementation, are robust to substantial violations of the parallel trends assumption.

Discussion

On the whole, our retrospective analysis suggests that the 2001 Arsenic Rule under the SDWA significantly reduced As occurrence among U.S. community water systems. Statistically significant decreases in average As occurrence begin as early as 2001 and continue through the last year of our sample, 2019. Reductions in As are largest for small systems and those relying exclusively on groundwater as a source, and they vary significantly by region.

Our approach is designed to identify the causal effect of the As Rule on As occurrence among U.S. community water systems. Our models compare differences pre- vs. post-Rule between systems for which the Rule was likely to bind (i.e., those that had exceeded the new MCL prior to its promulgation) and a set of control systems that had never exceeded the new MCL pre-policy. Remaining sources of confounding variation would have to affect only treated systems, and only post-policy, in order to materially affect our results. The results are robust to varying assumptions about As concentrations in samples recorded as non-detects, to considering only finished water samples, to dropping states that adopted

their own, more stringent rules during the treatment period, and to several other alternative specifications.

One remaining concern might be that treatment CWS could have adopted new treatment technologies for compliance with other rules passed under the SDWA around the time of the Arsenic Rule, and that those technologies could also affect As occurrence. Most of the other SDWA rules were enacted well before the effective date of the Arsenic Rule, however, so they are unlikely to have contributed to the reductions in As levels observed in our analysis. For example, activated alumina, a treatment used to remove fluoride, also adsorbs As,⁴⁰ but the Fluoride Rule was passed in 1986, so installation of treatment processes likely began well before the early 2000s when our data first show a decline in As. To our knowledge, other rules promulgated around the time of the Arsenic Rule would not cause systems to adopt technologies or processes that would reduce As.

Our work provides causal evidence that As dropped significantly due to the As Rule, though we leave the quantification of health benefits of these reductions for future work. Moreover, a retrospective analysis of the Rule's costs would allow for a re-evaluation of the EPA's prospective conclusion in its regulatory impact analysis that the Rule's benefits were unlikely to exceed its costs, particularly for small systems. The SDWA Six Year Review data that we use to estimate the Rule's effect on As occurrence also include information on the drinking water treatment technologies in place at CWS. Unlike the analytic sampling data, the EPA has not applied quality assurance evaluations to these treatment technology data in the SYRs.²⁴ While the existing literature suggests that the capital costs of compliance with the As Rule may have been lower than anticipated in the RIA,³⁹ determining whether updated compliance cost estimates from the six-year review or other sources can be used to comprehensively update EPA's prospective cost estimate is an important area for future research.

Acknowledgement

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Supporting Information Available

Includes additional results figures and tables.

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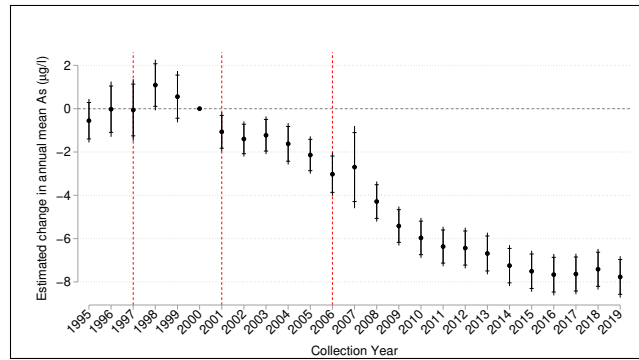
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TOC Graphic



IMPACTS OF THE U.S. SAFE DRINKING WATER ACT ARSENIC RULE ON ARSENIC OCCURRENCE

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1 Supporting Information

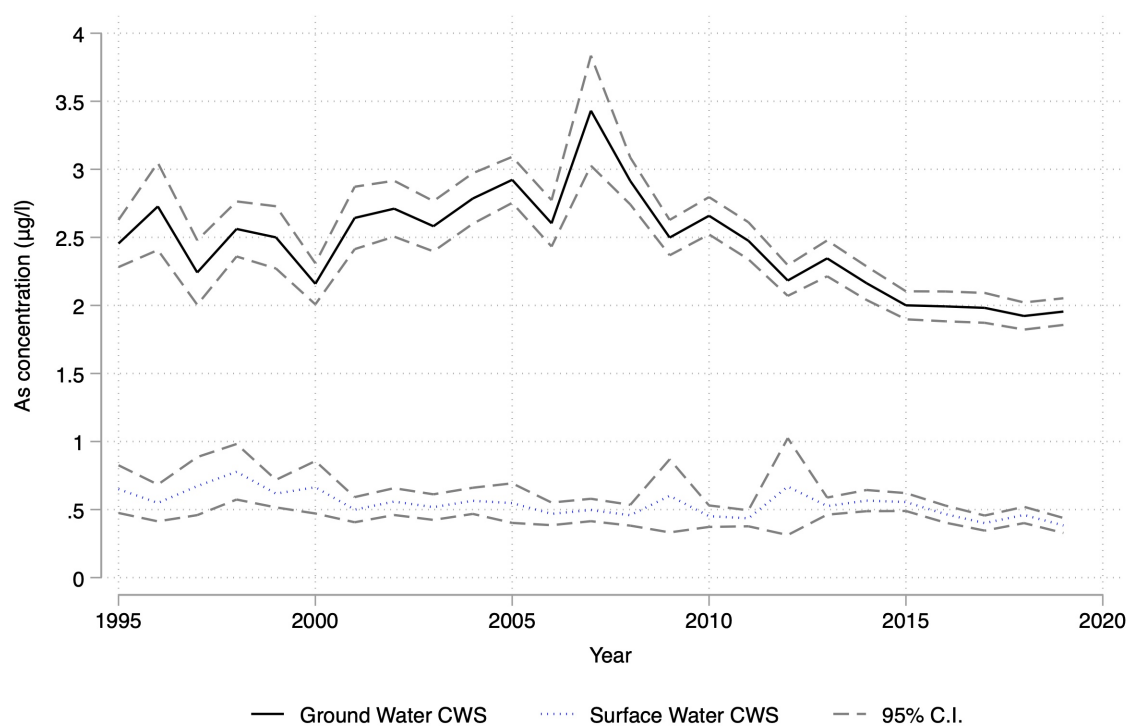
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Number of tables: 7

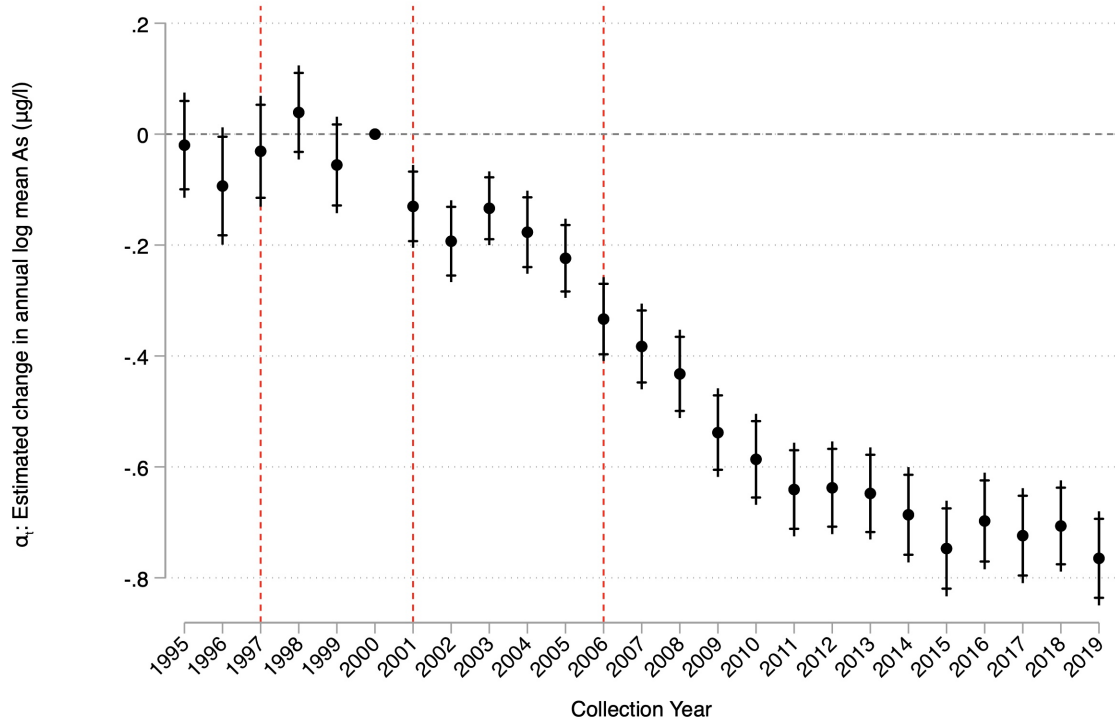
1.1 SI Figures

Figure S1: Trends in Average Annual Mean As Concentrations by Water Source, 1995 to 2019



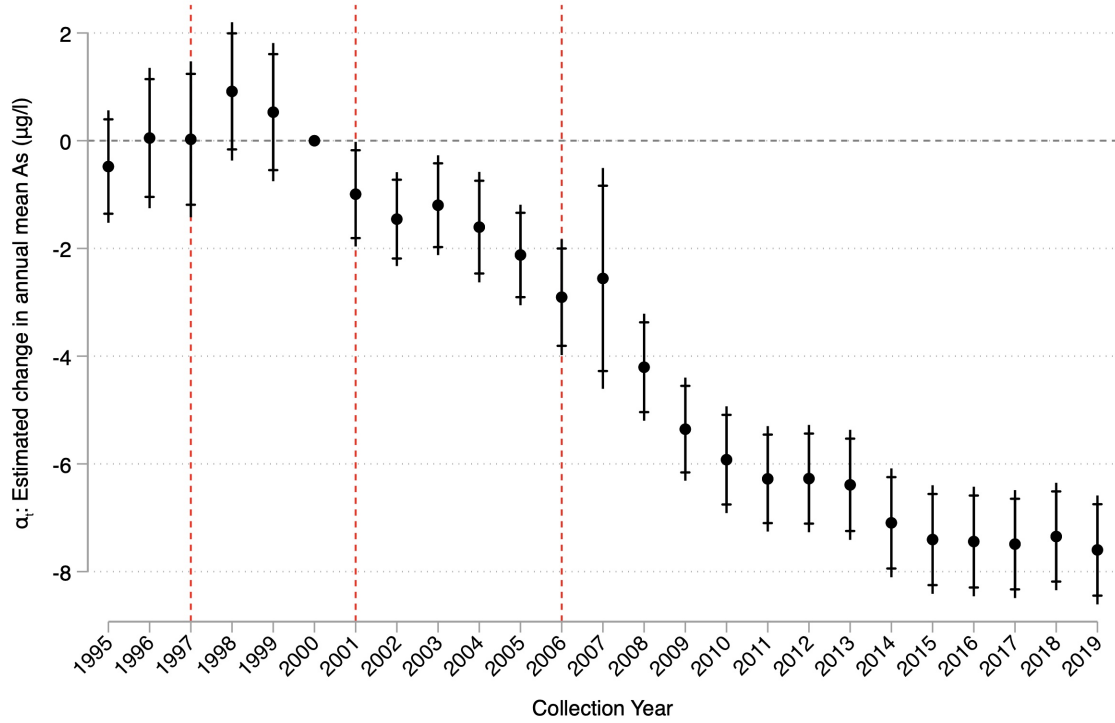
Notes: The figure graphs the annual average mean arsenic (As) concentration (and 95% confidence intervals) in $\mu\text{g/l}$ for the 22,127 community water systems (CWS) in our sample from 1995-2019 by water source. The panel is unbalanced, because we do not observe As concentrations for each system in every year.

Figure S2: Event Study Estimates with Log Specification



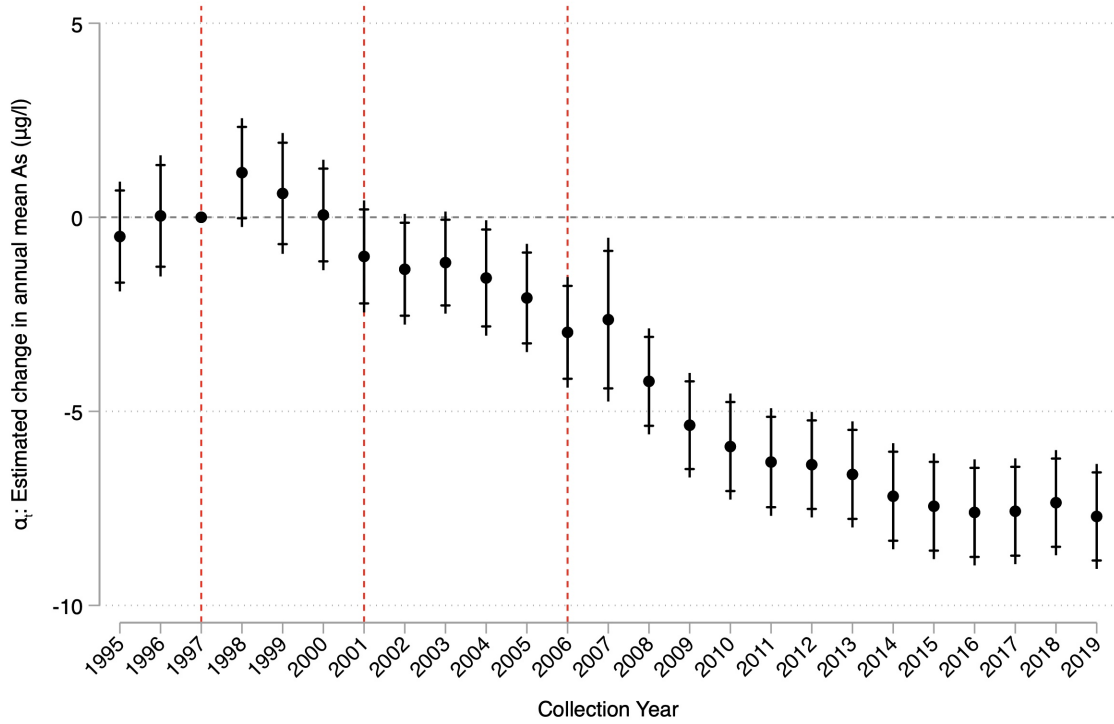
Notes: The figure shows estimated coefficients (α_t from Eq. 2, or the coefficients on the variable $treated_i \cdot D_{it}$) and 95% confidence intervals from an event study specification with log-transformed annual mean arsenic (As) concentrations as the dependent variable. Vertical lines in the figure represent the year before rule proposal (1996), year before rule promulgation (2000), and year of compliance (2006) with the rule. Coefficient estimates represent the average difference in the log mean annual As occurrence between treatment and control systems in each year, compared to 2000. The estimated coefficients in the post-treatment period (2001-2019) capture the causal effects of the Rule on As occurrence as they unfold over time. In the pre-treatment period (pre-2001), the coefficients are placebo tests; statistically non-zero estimates prior to the Rule would threaten the assumption of parallel trends in As occurrence between control and treatment water systems absent the Rule.

Figure S3: Event Study Results with Subsample of 24 States



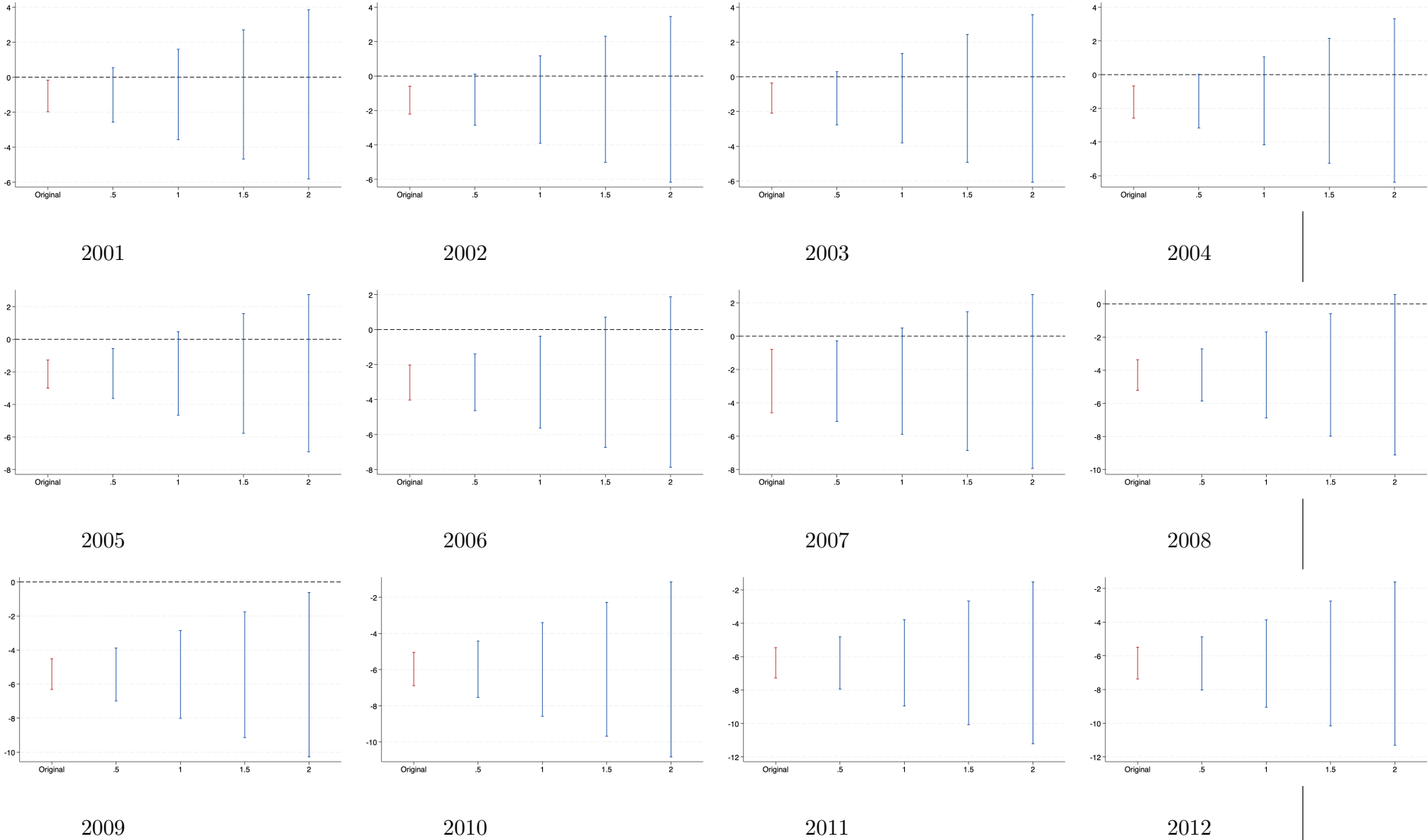
Notes: The figure shows estimated coefficients (α_t from Eq. 2, or the coefficients on the variable $treated_i \cdot D_{it}$) and 95% confidence intervals from an event study specification with annual mean arsenic (As) concentrations as the dependent variable. Vertical lines in the figure represent the year before rule proposal (1996), year before rule promulgation (2000), and year of compliance (2006) with the rule. The sample is restricted to community water systems located in the 24 states represented in the Rule occurrence data. Coefficient estimates represent the average difference in the mean annual As occurrence between treatment and control systems in each year, compared to 2000. The estimated coefficients in the post-treatment period (2001-2019) capture the causal effects of the Rule on As occurrence as they unfold over time. In the pre-treatment period (pre-2001), the coefficients are placebo tests; statistically non-zero estimates prior to the Rule would threaten the assumption of parallel trends in As occurrence between control and treatment water systems absent the Rule.

Figure S4: Event Study Results with Alternative Post-Treatment Period



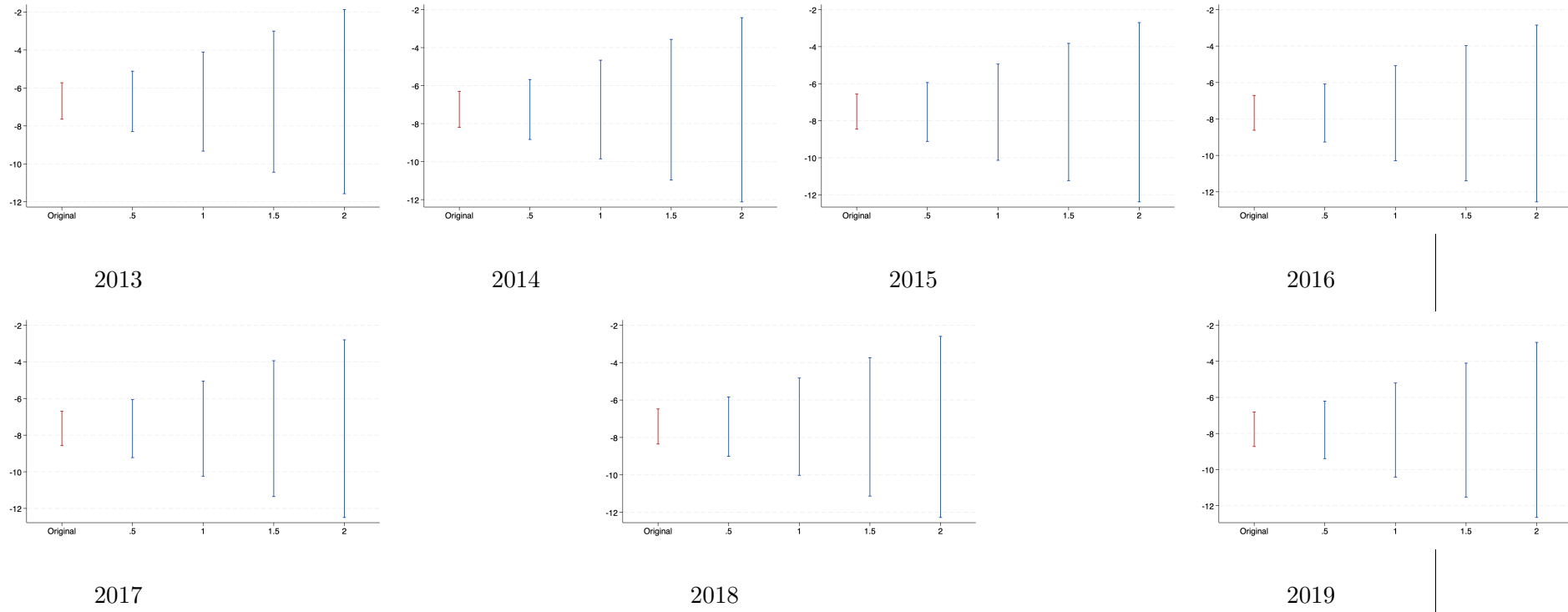
Notes: The figure shows estimated coefficients (α_t from Eq. 2, or the coefficients on the variable $treated_i \cdot D_{it}$) and 95% confidence intervals from an event study specification with annual mean arsenic (As) concentrations as the dependent variable. Vertical lines in the figure represent the year before rule proposal (1996), year before rule promulgation (2000), and year of compliance (2006) with the rule. Coefficient estimates represent the average difference in the mean annual As occurrence between treatment and control systems in each year, compared to 1997. The estimated coefficients in the post-treatment period (1998-2019) capture the causal effects of the Rule on As occurrence as they unfold over time. In the pre-treatment period (pre-1997), the coefficients are placebo tests; statistically non-zero estimates prior to the Rule would threaten the assumption of parallel trends in As occurrence between control and treatment water systems absent the Rule.

Figure S5: HonestDiD Results for Event Study Estimates, 2001–2012



Notes: Please see notes for [S6](#) below.

Figure S6: HonestDiD Results for Event Study Estimates, 2013–2019



Notes: Figure S5 and S6 show the HonestDiD sensitivity analyses for event study estimates in each post-treatment year, 2001-2019, compared to 2000. M values on the horizontal axis of each panel progressively relax the parallel trends assumption by allowing larger deviations between treatment and control systems post-treatment. For example, when M=2, treatment and control group trends could diverge by as much as twice the maximum deviation observed pre-treatment. Results for 2001-2005 suggest that our estimates of the Rule's impact on arsenic (As) concentrations are sensitive to violations of the parallel trends assumption. From 2006 onward (after the Rule's compliance deadline), coefficient estimates are statistically significant and of similar magnitude even with large post-treatment deviations, showing robustness to even substantial parallel trends violations.

1.2 SI Tables

Table S1: Sample vs. Population Balance Table

A. All CWS			
	Study CWS Mean/(SD) (1)	Non-Study CWS Mean/(SD) (2)	Pairwise t-test Mean difference (1)-(2)
Size: Very Small (Below 500)	0.53 (0.50)	0.58 (0.49)	-0.05***
Size: Small (501-3300)	0.27 (0.44)	0.26 (0.44)	0.01**
Size: Medium (3301-10000)	0.11 (0.31)	0.09 (0.28)	0.02***
Size: Large (10001-100000)	0.08 (0.28)	0.06 (0.25)	0.02***
Size: Very Large (Above 100000)	0.01 (0.10)	0.01 (0.08)	0.00***
Source: Ground Water	0.84 (0.36)	0.75 (0.44)	0.10***
Region: Northeast	0.09 (0.29)	0.15 (0.35)	-0.06***
Region: South	0.36 (0.48)	0.46 (0.50)	-0.10***
Region: Midwest	0.30 (0.46)	0.17 (0.37)	0.13***
Region: West	0.25 (0.43)	0.23 (0.42)	0.02***
B. Small or Very Small CWS Only			
	Study CWS Mean/(SD) (1)	Non-Study CWS Mean/(SD) (2)	Pairwise t-test Mean difference (1)-(2)
Source: Ground Water	0.92 (0.28)	0.81 (0.40)	0.11***
Region: Northeast	0.09 (0.29)	0.15 (0.35)	-0.06***
Region: South	0.35 (0.48)	0.44 (0.50)	-0.09***
Region: Midwest	0.31 (0.46)	0.16 (0.37)	0.14***
Region: West	0.25 (0.43)	0.25 (0.43)	0.00

Notes: The table reports the means of community water system (CWS) characteristics in the study sample versus the population. Standard deviations are in parentheses. There were 152,321 active public water systems in 2011 across the US and, of those, 51,338 were of the CWS type. Panel A of the table compares characteristics of all community water systems in the study sample (column 1) or all non-study CWS that are active as of 2011 (column 2). Column (3) reports the difference in means by treatment status and significance associated with a t-test for equality of means. 22,127 unique CWS are in the study sample and 30,243 unique CWS are in the non-study population. Panel B further limits the comparison to very small and small CWS only, since these groups see the largest effects from the policy. Among this group, 17,691 unique CWS are in the study sample and 25,527 unique CWS are in the non-study population. We note that there are 1,007 CWS that are in our study sample but recorded as inactive as of 2011. Characteristics are based on water system information in 2011 from the SDWIS. The system characteristics include the share of systems in various categories based on system size, water source, and geographic region. Geographical regions correspond to the four U.S. Census regions: Northeast, Midwest, West, and South.

Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S2: Balance Table

	Treatment Mean/(SD) (1)	Control Mean/(SD) (2)	Pairwise t-test Mean difference (1)-(2)
Size: Very Small (Below 500)	0.53 (0.50)	0.56 (0.50)	-0.03***
Size: Small (501-3300)	0.25 (0.43)	0.26 (0.44)	-0.01
Size: Medium (3301-10000)	0.10 (0.30)	0.09 (0.29)	0.01
Size: Large (10001-100000)	0.09 (0.29)	0.07 (0.25)	0.02***
Size: Very Large (Above 100000)	0.02 (0.14)	0.01 (0.09)	0.01***
Source: Ground Water	0.92 (0.27)	0.87 (0.34)	0.05***
Region: Northeast	0.08 (0.28)	0.09 (0.29)	-0.01
Region: South	0.22 (0.41)	0.37 (0.48)	-0.15***
Region: Midwest	0.22 (0.42)	0.31 (0.46)	-0.09***
Region: West	0.48 (0.50)	0.23 (0.42)	0.25***

Notes: The table reports

the means of community water system (CWS) characteristics by treatment status. Geographical regions correspond to the four U.S. Census regions: Northeast, Midwest, West, and South. Standard deviations are in parentheses. The final column reports the difference in means by treatment status and significance associated with a t-test for equality of means. 1,952 unique CWS are assigned to the treatment group. 20,175 unique CWS are assigned to the control group. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S3: Estimated Impact of the Arsenic Rule on Log Mean As Concentrations

	Post-treatment period	
	2001+ (1)	1997+ (2)
β_1	-0.5*** (0.0)	-0.4*** (0.0)
Estimated change in As concentration	-4.7	-3.7
Pre-treatment average	13.0	12.3
Effect size	-36.4%	-29.8%

Notes: The table presents the estimated impact of the Rule on log-transformed annual mean arsenic (As) concentrations during two post-treatment periods: 2001+ and 1997+. Standard errors are clustered at the water system level and are provided in parentheses. Estimated changes in annual mean As concentrations are calculated by exponentiating the coefficients and applying them to the pre-treatment average of mean As concentrations for the treatment group. Effect sizes are reported as percentage changes relative to the pre-treatment average. Pre-treatment average is the average of annual mean As concentration for treatment systems in the pre-treatment period, as defined in the respective column. Sample size is 64,624 observations, representing 10,127 unique community water systems. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table S4: Estimated As Rule Impact Excluding New Hampshire and New Jersey

Sample:	Exclude	
	Baseline	NJ and NH
	(1)	(2)
β_1	-5.0*** (0.3)	-4.8*** (0.3)
Obs.	192,347	184,035

Notes: The table presents the estimated impact of the Rule on annual mean arsenic (As) concentrations during the post-treatment period of 2001+ for two samples: the main estimation sample (“baseline”) and without water systems from New Jersey and New Hampshire. Standard errors are clustered at the water system level and are provided in parentheses.

Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Baseline sample size is 192,347 observations, representing 22,127 unique community water systems. Excluding the two states removes 868 unique CWS (402 from NH and 466 from NJ) from the estimation sample.

Table S5: Estimated As Rule Impact under Alternative Sampling Frequency Restrictions

Sampling Freq.	Every 3 yrs		Active
Restriction:	Baseline	2001-2019	1995-2019
	(1)	(2)	(3)
β_1	-5.0*** (0.3)	-5.3*** (0.4)	-5.4*** (0.4)
Obs.	192,347	129,811	120,879

Notes: The table presents the estimated impact of the Rule on annual mean arsenic (As) concentrations during the post-treatment period, 2001+. Column (1) is the main result (reproduced from Table 2 of the main text), column (2) limits the sample to systems with at least one sample every three years in the post-treatment period (2001-2019), column (3) limits the sample to systems with at least one sample every three years for the entire period (1995-2019), and column (4) limits the sample to systems that were active as of 2015. Standard errors are clustered at the water system level and are provided in parentheses. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Pre-treatment average is the average of annual mean As concentration for treatment systems in the pre-treatment period, as defined in the respective columns.

Table S6: Estimated As Rule Impact under Alternative Non-Detect Assumptions

	Zero	1/2 Detection	Detection		
Non-Detect Value:	(Baseline)	Limit	Limit	Zero	1.4 μ g/l
	(1)	(2)	(3)	(4)	
β_1	-5.0*** (0.3)	-6.9*** (0.3)	-8.9*** (0.4)	-4.6*** (0.3)	-4.8*** (0.3)
Obs.	192,347	160,645	160,645	160,645	192,347

Notes: The table presents the estimated impact of the Rule on annual mean arsenic (As) concentrations under the 2001+ post-treatment period with alternative imputation of the non-detect value. Column (1) presents baseline results, which imputes non-detects as zero, column (2) imputes non-detect values with 50% of the detection limit, and column (3) imputes non-detect values with the detection limit. Sample sizes vary since detection limits are missing in some cases. Column (4) uses the baseline imputation value of zero but with the sample of systems in column (2) and (3). Column (5) uses a detection limit of 1.4 μ g/l based on the method detection limit for Method 200.8. Standard errors are clustered at the water system level and are provided in parentheses. Statistical significance is denoted as follows: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sample size in the baseline is 192,347 observations, representing 22,127 unique community water systems. Sample size with alternative imputation values for non-detections in columns (2) and (3) are 160,645 representing 22,031 unique community water systems.

Table S7: Estimated As Rule Impact based on Raw or Finished Water

Data Source:	SYR2 Data for 1998-2000				Occurrence Data for 1998-2000
Sample Inclusion:	All (Baseline)	No Raw	Finished	Raw	Finished
	(1)	(2)	(3)	(4)	(5)
β_1 D	-5.0*** (0.3)	-5.7*** (0.3)	-5.5*** (0.4)	-1.8*** (0.6)	-6.2*** (0.5)
Obs.	192,347	171,161	153,931	14,885	71,003

The table presents the estimated impact of the Rule on annual mean arsenic (As) concentrations under the 2001+ post-treatment period using various samples based on the type of water sampling (finished or raw). Column (1) presents the main results using all of the sampling data, regardless of the type of water sample. Sample size in the baseline is 192,347 observations, representing 22,127 unique community water systems. Column (2) presents results that drops any identifiable raw water samples (the robustness check in the original submission), which includes 20,494 unique water systems across 44 states. Column (3) keeps only identifiable finished water samples, which limits the sample to 17,990 unique water systems across 43 states. Column (4) includes identifiable raw water samples only, which limits the sample to 1,679 unique systems from 13 states. Columns (1)-(4) rely on sampling data from SYR2 for the years 1998-2000. Finally, column (5) includes finished water samples only and uses the occurrence data to capture sampling in 1998-2000, limiting the sample to 9,970 unique water systems across 19 states. This sample removes (5) outlier observations with As concentrations of greater than 3,400.